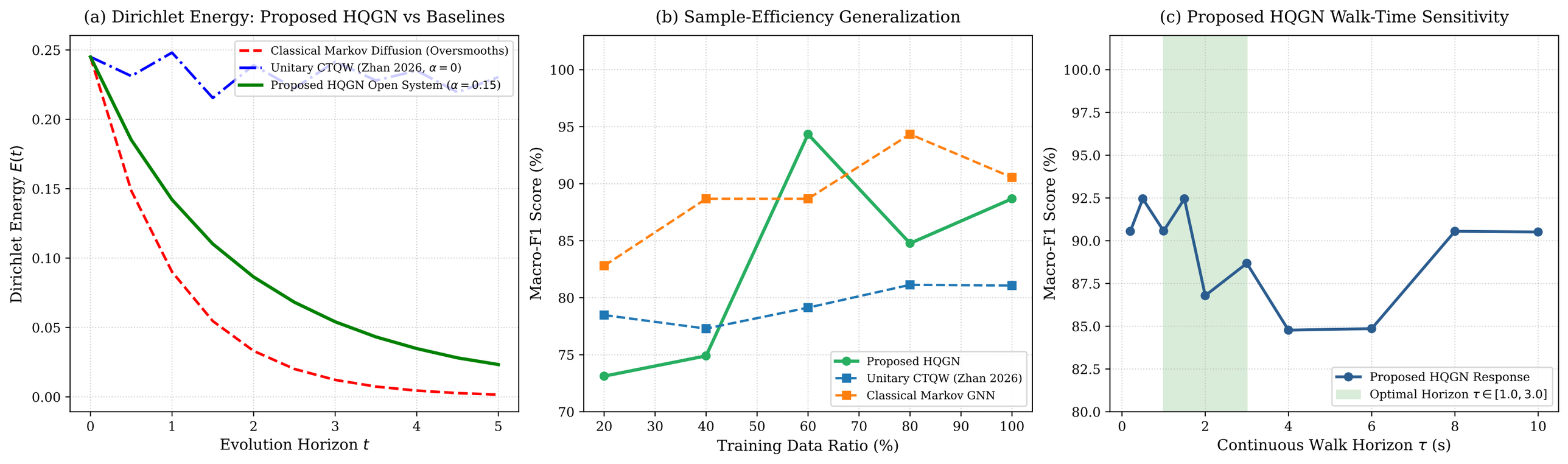
(a) Dirichlet Energy: Proposed HQGN vs Baselines
Classical Markov Diffusion (Oversmooths)
Unitary CTQW (Zhan 2026, $\alpha = 0$)
Proposed HQGN Open System ($\alpha = 0.15$)
Dirichlet Energy $E(t)$
Evolution Horizon t
(b) Sample-Efficiency Generalization
Proposed HQGN
Unitary CTQW (Zhan 2026)
Classical Markov GNN
Macro-F1 Score (%)
Training Data Ratio (%)
(c) Proposed HQGN Walk-Time Sensitivity
Proposed HQGN Response
Optimal Horizon $\tau \in [1.0, 3.0]$
Macro-F1 Score (%)
Continuous Walk Horizon τ (s)